An alternative proposal for dynamic ISF in open-source AID systems

The Diabeloop glucose curve, tested on other cohorts and set against the logarithmic and sigmoid models

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Summary

The dynamic ISF models in AndroidAPS and Trio have two origins. The logarithmic model is Chris Wilson's. It was built from a volunteer survey group of 50 people using multiple daily injections, who supplied the manual corrections they gave and the outcomes, with glucose from fingerpricks and CGM. The sigmoid model offered in Trio came from its author's observation of glucose response, and no analysis of data has been published in support of it.

Diabeloop has since examined the same idea on the records of its own closed-loop system and presented the result at the American Diabetes Association's Scientific Sessions in 2025 [1]. From 11,201 adults it reported a sensitivity of 79 mg/dL per unit at a glucose of 100 mg/dL, 35 at 180 and below 20 above 300, with a fitted equation.

This document builds on that work. It sets out the equation and how it was derived, and reports what the same measurement shows on other cohorts. It then describes how the equation could be offered in AndroidAPS and Trio as an alternative to both the logarithmic and the sigmoid models.

The measurement was repeated on 1,790 people in eight cohorts held in this series. Within person, the value at 100 mg/dL was 2.2 to 4.0 times the value at 200, against 2.61 in the published equation, so the steep fall towards 200 mg/dL is reproducible. On records where no algorithm was dosing, the published equation fits the measured values from 100 to 200 mg/dL at a single scale. On Loop records no single scale fits. Refitting the equation's five

coefficients to each cohort agreed with both results, and returned a flatter curve than the published one above 200 mg/dL. Simulated people whose sensitivity was constant also returned a falling curve by this method, so the decline between 150 and 250 mg/dL is not evidence that a unit of insulin does less at higher glucose.

The published equation has no term for the person, and it needs one before it can dose. Two are described here: total daily dose (TDD), as the logarithmic model uses, and the profile ISF. For an adult on 40 U a day with a profile ISF of 50 mg/dL per unit, the equation scaled by TDD gives 109.4 mg/dL per unit at 100 mg/dL and 41.9 at 200. AndroidAPS at an adjustment factor of 100% gives that person 43.4 and 29.3. Both scalings are modelling choices made in this document and appear nowhere in the publication.

Two parts of the model are unsettled. The first is its level, since the publication reports no daily dose and the candidates in this series differ by more than a factor of four. The second is where on the curve a correction is read, which moves the dose by 31 to 49%. Whether a loop dosing to this equation does better than one dosing to either existing model has not been tested.

This is a draft for discussion. The equation has not run on a device or in a closed-loop simulator, and the candidate uses the published coefficients unchanged. Readers are assumed to know how dynamic ISF works in both systems. Glucose is in mg/dL throughout because the code is.

What is the Diabeloop equation?

The source [1] is a conference abstract and poster from Diabeloop, the maker of the DBLG1 closed-loop system. It used the records of 11,201 adults with type 1 diabetes from March 2021 onwards. No full paper was found.

The abstract reports a sensitivity that rises from 65 to 79 mg/dL per unit between 50 and 100 mg/dL. Above that it falls, to 35 at 180 mg/dL and below 20 above 300. Relative to a dose sized at 180 mg/dL, the authors suggest the dose could be halved at 120 mg/dL or below and may need doubling above 340.

The poster's results figure gives the fit in two pieces, for glucose G in mg/dL. Up to 100 mg/dL:

```
ISF = 98.03 − 1.077 G + 0.008868 G^2
```

Above 100 and up to 380 mg/dL:

```
ISF = 272 − 3.121 G + 0.01511 G^2 − 3.305e−05 G^3 + 2.69e−08 G^4
```

The coefficients in the code (`inv008.dynisf.quartic_isf`) were checked against that figure on 11 November 2026 and match digit for digit. The quartic gives 80.6 mg/dL per unit at 100 mg/dL, 35.3 at 180, 21.1 at 300 and 15.3 at 380. It implies dose multiples of 0.56 at 120 mg/dL and 1.95 at 340, in line with the abstract's half and double. The quadratic gives 79.0 at 100 mg/dL and 65.3 at 60, so the two pieces differ by 2% where they meet.

The equation is steeper than the logarithmic model. Between 100 and 200 mg/dL its value falls by a factor of 2.61, where the logarithmic model with a divisor of 55 falls by 1.48. From 200 to 300 mg/dL its value is multiplied by 0.69 and the logarithmic model's by 0.82.

It has no term for the person. Everyone gets 35.3 mg/dL per unit at 180 mg/dL, whatever their daily dose or their profile.

How was it derived?

The unit of analysis is a fall in glucose. On days with at least 75% of time in closed loop, the authors took each glucose peak between 50 and 400 mg/dL that came at least an hour after a meal, with less than 2.5 g of carbohydrate still on board, and followed it to the next minimum more than 20 minutes later. The sensitivity for the event is the fall divided by the insulin used over it. Insulin used is the drop in insulin on board plus whatever was delivered above the reference basal. The curve is the fit through those event values against glucose.

Each event ends where glucose stopped falling, so events are selected on what glucose did. The figure is therefore a ratio of two outcomes, the fall and the insulin used during it. That bears on how the curve is read in the next section.

Four things the derivation depends on are absent from the publication. It does not define the reference basal or say which insulin action model produced the insulin on board. The figure's horizontal axis is labelled G(t) with no statement of which glucose in an event that is. It gives no daily dose for the people measured. The rest of the poster could not be retrieved, and the authors have not been asked.

What does it show on our data?

The same construction was run in this series as INV-012, on 1,790 people in eight cohorts, and is reported in section 7 of document 5. It was built before the abstract was read, and two of its event rules differ from the published ones. It required three hours clear of a meal where the publication requires one hour with under 2.5 g on board, and a peak standing 15 mg/dL clear of its surroundings where the publication requires a 20 minute interval. Each event was filed under the glucose halfway between its peak and its minimum, and multiplied by the person's daily dose.

The shape between 100 and 200 mg/dL

The steep fall is reproducible. Within person, the value at 100 mg/dL was 2.2 to 4.0 times the value at 200 mg/dL across the eight cohorts, against 2.61 in the published equation. In REPLACE-BG, where no algorithm was dosing, the ratio was 2.55 (95% interval 2.25 to 2.82), which includes the published figure. Under closed loop it was steeper: 3.26 (3.11 to 3.39) for Loop, 3.86 (2.72 to 6.04) for oref0 and 3.21 (2.24 to 5.62) for Trio. The first two intervals lie above 2.61 and Trio's includes it.

Fitting the equation to the measured values

The table divides the measured value in each glucose bin by the published value at that glucose. The result is the daily dose, in U, at which the published equation would return what was measured. A row that stays level means the published shape fits that cohort at one scale. Only the four cohorts with carbohydrate records are shown, with the 95% interval over people.

Cohort	People	100 mg/dL	150	200	250	300
No algorithm (REPLACE-BG)	196	103 (89 to 115)	100 (94 to 108)	106 (94 to 125)	120 (103 to 137)	174 (132 to 233)
Loop	842	88 (84 to 91)	58 (56 to 60)	71 (67 to 73)	82 (77 to 89)	108 (95 to 119)
Trio	22	90 (63 to 111)	52 (40 to 64)	74 (51 to 104)		
oref0	86	109 (90 to 136)	70 (59 to 86)	78 (58 to 188)		

With no algorithm dosing, the published equation fits from 100 to 200 mg/dL at one scale: 103, 100 and 106 U, each inside the other two intervals. Above 200 mg/dL the row rises to 120 and then 174 U, so the published equation

falls faster there than these records do. Under Loop no single scale fits. The figure is 88 U at 100 mg/dL, 58 at 150 and 71 at 200, and those three intervals do not overlap. Trio and oref0 show the same dip at 150 mg/dL with wider intervals.

Two cautions apply to the table. The columns hold different people, since fewer reach the higher bins, and the within-person ratios in this section are the firmer comparison. The published values were read at the centre of each bin on the assumption that the publication files an event under its midpoint glucose, which it does not state.

The scale itself is not a daily dose anyone was using. It is about 100 U in REPLACE-BG because the measurement reads high, as the simulation below shows.

Refitting the equation

The equation was also refitted. The quartic, with all five coefficients free, was fitted to each cohort's values in 10 mg/dL bins, from 100 mg/dL to the highest bin holding ten people. Each bin was weighted by the square root of the people in it, which is a modelling choice, since the poster does not state its weighting. For comparison the published fit was laid over the same bins with one free number, its scale in U. The miss is the typical distance between a fit and the bin values, as a percentage of each value.

Cohort	People	Bins fitted, mg/dL	Best scale of the published fit, U	Miss, published fit	Miss, refit
No algorithm (REPLACE-BG)	196	100 to 370	103 (96 to 113)	14%	7%
Loop	842	100 to 380	71 (69 to 74)	17%	6%
Trio	22	100 to 200	65 (48 to 80)	18%	7%
oref0	86	100 to 240	82 (71 to 99)	23%	14%

The refit misses by less in every cohort, as a fit with four more free numbers must. The second table shows where the two differ. It gives each fit relative to its own value at 180 mg/dL.

Fit	People	100 mg/dL	150	200	250	300
Published		2.29	1.30	0.87	0.70	0.60
Refit, no algorithm	196	2.17 (1.91 to 2.40)	1.28 (1.21 to 1.33)	0.89 (0.86 to 0.92)	0.81 (0.72 to 0.87)	0.90 (0.77 to 1.02)

Fit (REPLACE-BG)	People	100 mg/dL	150	200	250	300
Refit, Loop	842	3.19 (3.02 to 3.34)	1.25 (1.22 to 1.27)	1.00 (0.98 to 1.01)	1.07 (1.03 to 1.11)	0.94 (0.88 to 1.01)
Refit, Trio	22	3.29 (2.15 to 4.76)	1.19 (0.85 to 1.53)	1.12 (0.66 to 1.61)		
Refit, oref0	86	2.96 (2.22 to 3.83)	1.00 (0.78 to 1.24)	1.02 (0.82 to 1.38)		

With no algorithm dosing, the refit returns the published shape from 100 to 200 mg/dL, and all three intervals there include the published figure. At 250 and 300 mg/dL the refit sits at 0.81 and 0.90 of its value at 180, where the published fit is at 0.70 and 0.60. Both of those intervals exclude the published figure. Under Loop the refit differs from the published fit at every glucose in the table. It is higher at 100 mg/dL, 3.19 against 2.29, and from 200 mg/dL upward it stays close to its value at 180. Trio's intervals, from 22 people, include the published figure at each of its three points. The oref0 values resemble Loop's, with an interval that excludes the published figure only at 150 mg/dL.

Two limits apply. A quartic suits these records poorly, because the values fall steeply to about 150 mg/dL and then level off. Under Loop the refit rises from 1.00 to 1.07 between 200 and 250 mg/dL, while the bin values themselves go from 2,182 to 2,025. Above 200 mg/dL the bins and the within-person ratios reported below are the better guide. The second limit is that the refit describes the same outcome ratio as the published curve, so the simulation in the next subsection bears on it equally.

The refit gives no single equation to offer in place of the published one. Its coefficients differ between cohorts, and its value at 180 mg/dL is 3,657 (3,431 to 4,055) with no algorithm dosing and 2,123 (2,048 to 2,214) under Loop. It confirms the published shape between 100 and 200 mg/dL on records that no loop shaped. It also agrees with the within-person ratios that, by this measure, the published fall above 200 mg/dL is not reproduced here.

What a constant sensitivity returns

A constant sensitivity produces a falling curve by the same method. Simulated people with a true value of 2,000 at every glucose returned values 1.4 to 3.4 times too high, declining as glucose rose. Their ratio of the value at 250 to the value at 200 mg/dL was 0.87 (0.83 to 0.90) in one arm and 0.92 (0.89 to 0.95) in the other. The six Jaeb cohorts gave 0.84 to 1.05, a range

that contains both simulated arms, so the decline in those cohorts between 150 and 250 mg/dL needs no change in what a unit of insulin does to explain it. The published equation gives 0.80, which is steeper than either simulated arm by an amount this series cannot account for. The simulation produced no usable events at 100 or at 300 mg/dL and says nothing about either end.

Near target

Near target the decline comes from the insulin, with the fall almost unchanged. Across cohorts the median fall was 30 to 43 mg/dL for events at 100 mg/dL and 39 to 64 for events at 200. The median insulin used was 0.02 to 0.2 U at 100 mg/dL and 0.3 to 1.5 U at 200. A similar fall divided by much less insulin gives a much higher figure. At 100 mg/dL between 9 and 58% of events had no net insulin used at all, and a line fitted through each person's events put the fall at zero net insulin at 27 to 44 mg/dL. In type 1 diabetes a raised glucose does not fall without insulin, so that fall is basal and earlier doses acting, together with the selection of events that end at a minimum. The steep part of the curve below 150 mg/dL therefore rests on very small net doses, and it has not been tested against a known truth.

Above 200 mg/dL

The two measures disagree here. By the ratio of fall to insulin used, the value at 300 against 200 mg/dL within person was 0.90 (0.77 to 1.26) in REPLACE-BG, from 60 people, and 0.96 (0.84 to 1.08) under Loop, from 137 people. Both intervals exclude the published 0.69. Measured as the extra fall per extra unit, the same comparison gave 0.66 (0.49 to 0.73) in REPLACE-BG, from 24 people, and 0.73 (0.64 to 0.93) under Loop, from 49 people, and both of those include 0.69. No simulated arm reached 300 mg/dL, so neither measure has a known-truth check there.

As a predictor

The shape has been tried once as a predictor on oref records, where it did worse than the static profile. On 80 people and 62,344 overnight windows, with each shape anchored to the profile at 100 mg/dL, the typical miss was 24.2 mg/dL for the static profile and 29.8 for this shape. At 205 to 260 mg/dL it was 37.8 against 73.7. No interval was reported for those figures. Two things limit the test. Its windows were selected on the later outcome, as the audit of 16 September 2026 recorded, and it anchored the shape at 100 mg/dL where this document anchors at 180.

Where that leaves the equation

All of this is observational, and the publication is a conference abstract from the system's manufacturer. As a description of net fall per net unit in events chosen from peak to minimum, the equation is reproducible on other people's data between 100 and 200 mg/dL. As a statement that a unit of insulin does less at higher glucose, it is unsupported between 150 and 250 mg/dL and open on either side of that range. Whether a loop that doses to it does better than one dosing to either existing model or to a flat profile is a separate question. No evidence in this series bears on it directly.

A proposal from June 2026 that used this shape (document 3) was withdrawn for live dosing on 16 September 2026. The audit of that date left the way open for such candidates to be simulated and replayed in shadow, and to be studied prospectively. The evaluation proposed below stays inside that.

How could it be implemented?

What it would replace

AndroidAPS and Trio run the same logarithmic equation:

sensitivity = 1800 ÷ (adjustment factor × TDD × ln(glucose ÷ divisor + 1))

Trio computes a ratio, profile ISF × adjustment factor × TDD × ln(glucose ÷ divisor + 1) ÷ 1800, and divides the profile ISF by it, so the profile value cancels. Given the same inputs the two systems return the same number. This was checked on 11 November 2026 against AndroidAPS dev 0c593a4b09 and Trio dev 902be75e8, both of 10 November 2026. Neither calculation differs from the released versions (AndroidAPS master f77abd9b82, Trio main 0b29cd534), and Trio's `DynamicISF.swift` was last changed on 24 August 2026.

What separates the two in use is the defaults around the equation. Trio's adjustment factor defaults to 0.8 where AndroidAPS defaults to 100%, which puts Trio's value 25% higher. Trio then holds its ratio inside the autosens limits, 0.7 and 1.2 by default, and `calculateRawDynIsf` in AndroidAPS applies no limit. For the worked person Trio's upper limit binds from 157 mg/dL. Above that glucose its value stays at 41.7 mg/dL per unit while AndroidAPS continues down to 29.3 at 200 mg/dL. The daily dose is blended

differently in each system, and the divisors differ for faster insulins: 55, 65 or 75 by peak time in AndroidAPS, and 55 or 70 in Trio, or 120 less a custom peak time.

Trio's sigmoid mode returns the profile ISF at the profile target and moves it within the same limits. oref0 as published has autosens and no dynamic ISF based on daily dose. Loop and iAPS were not read.

Linked to TDD, or to the profile ISF?

The published values need a scale for each person. At 180 mg/dL the equation gives 35.3 mg/dL per unit to a person on 20 U a day and to a person on 80 U a day. Among 422 adults with entered pump settings analysed earlier in this series, sensitivity multiplied by daily dose was 1915 (95% CI 1828 to 1986), which puts the first of those people near 96 mg/dL per unit and the second near 24.

The publication's own reference point is 180 mg/dL, the glucose against which it states its halving and doubling of doses. Both links below hold a value at that glucose and keep the published shape on either side of it.

Linked to TDD, a person's sensitivity at 180 mg/dL is taken to be 1915 ÷ TDD:

sensitivity = (1915 ÷ TDD) × (published value at this glucose ÷ published value at 180 mg/dL)

This is the logarithmic model with its glucose term swapped. The daily dose is each system's existing blend and the adjustment factor multiplies it as now. The divisor for insulin type has no counterpart and drops out. The value keeps moving with recent insulin use by the same multiplier as today: 1.39 for a person whose blended dose is 0.72 of their own average and 0.88 at 1.14 of it, which are the tenth and ninetieth percentiles among 960 people in document 6. The profile ISF and its schedule through the day play no part, as in AndroidAPS now. The adult figure of 1915 is a modelling choice. The figure pooled across all ages in the same analysis is higher, and 51% of that group are children, so it was not used.

Linked to the profile, the entered ISF is taken to hold at 180 mg/dL:

sensitivity = profile ISF × (published value at this glucose ÷ published value at 180 mg/dL)

This fits Trio's structure, which already expresses dynamic ISF as a ratio of the profile. The schedule through the day is kept and no level has to be chosen. The value stops responding to recent dosing. It is also only as good as the entered number. It suits a profile ISF the person has evaluated, and it carries the whole of any error in one they have not.

The two links agree for anyone whose profile ISF multiplied by daily dose is 1915. The worked person has a profile ISF of 50 mg/dL per unit, a daily dose of 40 U, rapid-acting insulin and recent dosing equal to their average. Their product is 2,000, which is why their two sets of values sit about 4% apart. The table gives the sensitivity each method assigns them, in mg/dL per unit, with both systems at their defaults.

Glucose, mg/dL	AndroidAPS, divisor 55	Trio logarithmic	Trio sigmoid	Equation linked to TDD	Equation linked to profile
80	50.1	62.6	53.7	109.4	114.3
100	43.4	54.3	50.0	109.4	114.3
120	38.9	48.6	47.1	86.2	90.1
150	34.2	42.8	44.3	62.3	65.1
180	31.0	41.7	42.9	47.9	50.0
200	29.3	41.7	42.4	41.9	43.7
250	26.3	41.7	41.8	33.7	35.1
300	24.1	41.7	41.7	28.7	30.0
380	21.8	41.7	41.7	20.7	21.7

Against AndroidAPS at 100% the equation linked to TDD gives less insulin at every glucose up to 362 mg/dL with a divisor of 55, and up to 313 mg/dL with a divisor of 75. The ratio between the two is 2.52 at 100 mg/dL, 1.82 at 150, 1.43 at 200 and 1.19 at 300. Against Trio at its defaults, in either mode, the equation gives less insulin below about 200 mg/dL and more above. Against this person's own profile it gives less insulin below 174 mg/dL and more above.

The published fit stops at both ends, and the reference implementation (`inv013/candidate.py`) makes a choice at each. Below 100 mg/dL it holds the value reached at 100. The published quadratic falls there, towards 65 mg/dL per unit at 60 mg/dL, and following it would give more insulin as glucose drops below 100. Work in this series in March 2026 declined the quadratic for dosing on that ground. Above 380 mg/dL, where the fit ends, the value at 380 is held.

This document carries the link to TDD as the candidate, because the logarithmic model is defined that way, and the link to the profile as an arm of the evaluation.

The level

The link to TDD needs a level, and it is the least settled part of the model because the publication gives nothing to set it from. One way to compare the candidates is the daily dose at which the published values would apply unchanged, called the reference dose here.

Basis for the level	People	Sensitivity × daily dose at 200 mg/dL (95% interval)	Reference dose, U
Adult entered settings, taken to hold at 180 mg/dL (this draft)	422	1,674 (1,598 to 1,736)	54.3
The 1800 rule, taken to hold at 105 mg/dL (this series, March 2026)		731	23.7
Measured under Loop	842	2,182 (2,065 to 2,266)	70.8
Measured under Trio	22	2,273 (1,569 to 3,218)	73.7
Measured under oref0	86	2,398 (1,779 to 5,809)	77.8
Measured with no algorithm dosing (REPLACE-BG)	196	3,283 (2,898 to 3,854)	106.5

The first row takes a person’s entered sensitivity to be right at 180 mg/dL. The second takes the conventional rule to be right near target, and it puts the whole curve at less than half the height of the first. The last four rows are the measured values from the previous section, read at 200 mg/dL. Trio’s interval, from 22 people, includes this draft’s 1,674. The other three intervals lie above it. The measured rows probably sit too high, since the same measurement on simulated people with a true value of 2,000 read 1.4 to 3.4 times above it, and daily dose was not defined the same way in each source.

No default can be chosen from these figures. The reference doses run from 24 to 107 U, and narrowing that range is the purpose of the second stage of the evaluation. For any one person the adjustment factor moves the level in either direction, as it does now. Its guidance has to be worked out again, because at 100% the new value at 100 mg/dL is 2.52 times the existing one.

Where on the curve a correction is read

A correction from 200 mg/dL to a target of 100 passes through every glucose in between, and the equation gives a different sensitivity at each. The existing code reads the logarithmic model at the current glucose, or at a glucose chosen by one of three branches in `DetermineBasalSMB.kt`. This draft assumes the publication filed each event under its midpoint glucose, and that assumption has not been confirmed.

The table gives the units needed to bring the worked person to 100 mg/dL under each reading. It is the arithmetic of the sensitivity alone, before insulin on board and the loop's predictions change it.

Starting glucose, mg/dL	AndroidAPS, divisor 55	Trio logarithmic	Profile	Equation, read at the start	Equation, read at the midpoint	Equation, summed along the path
150	1.46	1.17	1.00	0.80	0.61	0.62
200	3.41	2.40	2.00	2.39	1.61	1.62
250	5.71	3.60	3.00	4.46	3.01	2.97
300	8.29	4.80	4.00	6.97	4.78	4.58

Reading the equation at the starting glucose asks for 31 to 49% more insulin than reading it at the midpoint. From 200 mg/dL the two readings give 2.39 and 1.61 U, on either side of the 2.00 U the flat profile gives. Adding up the equation along the whole path gives 1.62 U. The midpoint reading stays within 0.04 U of that sum up to a starting glucose of 250 mg/dL and is 0.20 U above it at 300.

The default proposed is the midpoint between the glucose being corrected and the target, which is a modelling choice. Reading at the starting glucose is kept as an arm of the evaluation.

What changes in the code

The glucose function is the smallest part. One function per system returns the published quartic divided by its value at 180 mg/dL, with test vectors shared between the two codebases so that both return the same values.

Trio's limits need a decision from its maintainers. For the worked person the equation runs from 20.7 to 109.4 mg/dL per unit, and Trio's default limits would hold it between 41.7 and 71.4. That leaves it free to move only between 137 and 201 mg/dL, so with the limits unchanged most of the curve would never reach the dose.

The forecasts are affected as well as the dose. Both systems calculate a sensitivity at each predicted glucose, and the call sites can stay as they are. The equation's value rises 2.61 times as predicted glucose falls from 200 to 100 mg/dL, against 1.48 for the logarithmic model, so a forecast heading towards target is computed with a higher sensitivity than now. The effect on minimum predicted glucose and on time spent at zero basal has not been measured and is part of the replay.

The remaining additions are smaller. The loop result already stores the sensitivity it used (`variableSens` in AndroidAPS), and during evaluation both the existing and the new value are stored while the existing one doses. Settings gain a mode selector and, in a first version, no new number for the person to set. Each system gets unit tests on the shared reference vectors and replay assets for the new mode, alongside its existing tests (`ReplayApsResultsTest` in AndroidAPS, `DynamicISFTests` in Trio). A guard test confirms that the existing modes' output is unchanged.

In AndroidAPS the code touched is `calculateRawDynIsf` and the preference screen, together with the per-prediction lines and the dose-sensitivity branches in `DetermineBasalSMB.kt`. In Trio it is `DynamicISF.swift` and the forecast adjustment used in logarithmic mode.

Evaluation

Eight arms are compared throughout. Three are references. The logarithmic model as shipped and Trio's sigmoid are the two existing models, and the profile alone is the baseline for both. Four arms are forms of the equation: linked to TDD and read at the midpoint, the same read at the starting glucose, the same on a plain average daily dose, and linked to the profile. The eighth is

the first draft of this document, a sensitivity anchored to the profile and flat above 100 mg/dL. It is the arm the equation has to beat for its shape to matter.

The first stage replays each arm's value on recorded histories using the INV-008 pipeline. Results are split by glucose and by time since the last meal, and reported separately for people who announce meals and people who do not. The stage shows how much insulin each arm asks for and in whom. It is descriptive and has no pass rule.

The second stage runs the publication's measurement (`inv012/peak_trough.py`) on AndroidAPS and Trio records under the published event rules, and extends the simulated constant-sensitivity arms to reach 100 and 300 mg/dL. It decides the default level. It also shows whether the decline near target and above 250 mg/dL appears with no real effect behind it, as it does in between.

The third stage is simulation in TimSim, with runs paired by seed and arms for unannounced meals and for a mis-set adjustment factor. A harm rule is written before the runs and has two sides, time below 54 mg/dL and time above 250 mg/dL. The second side is there because this model gives less insulin than AndroidAPS at 100% over most of the range.

The fourth stage is shadow running. Both values are calculated on the device each cycle and stored, and the existing one doses. That gives the distribution of the ratio between them for each person, and from it a starting adjustment factor.

The fifth stage is use, on the developer's own device first and on released lines after that. It follows only when the third and fourth stages have passed, and it stops on the harm rule.

The first two stages are analysis and carry the analysis checklist. The last two change code that can change a dose. They carry unit and guard tests, the repository's review checklist, a replay against real history and a fresh-context review of the diff.

The way back at every stage is the mode selector. The existing modes stay in the code unchanged and remain the default until the fifth stage has run, and any build that includes the new mode names the build before it.

What is not settled

Some of what is missing can only come from the authors. The level depends on four things the publication does not give: which glucose an event is filed under, how the reference basal was defined, which insulin action model gave the insulin on board, and what daily doses the people measured were using. The authors have not been asked. They have also not been asked whether an equation derived by a manufacturer from its own users' records may be built into open-source code.

The level is open, as described above. A level set from entered settings at 180 mg/dL may not apply to open-source systems, since the measurement on Loop and oref0 records read higher at 200 mg/dL and only Trio's wide interval includes it.

It is an open question whether the steep part between 100 and 150 mg/dL should be used for dosing at all, given how many events there involved no net insulin. The question of whether to use the published quadratic below 100 mg/dL belongs with it.

The fall above 200 mg/dL is open in the same way. By the ratio of fall to insulin used, neither REPLACE-BG nor Loop reproduces it, whichever way that ratio is summarised. By the extra fall per extra unit, both do.

Parts of the existing systems were not read. The bolus calculator's sensitivity in dynamic mode is one, and the behaviour of autosens in that mode is another. TimSim's oref adapter has not been checked for whether it runs the logarithmic model closely enough to act as the comparator in the third stage.

The last question is for the maintainers of AndroidAPS and Trio, and it is whether they want one shared definition. The two systems run the same equation and, at their defaults, return values 42% apart for the worked person at 200 mg/dL. A reference implementation with shared test vectors would be new to both.

Reproducing the numbers

`python3 -m inv013.candidate` writes `results/inv013_comparator.md`, which holds the numbers in this document's worked examples and in every table except the refit. `python3 -m inv013.refit` writes `results/inv013_refit.md`, with the refit tables and the refitted coefficients. It reads the per-person bins that `python3 -m inv012.peak_trough` writes, a file too large to keep in git, and checks them against the committed INV-012 table

before fitting. `inv013/tests/test_candidate.py` checks the logarithmic equation against the replay code of INV-008, Trio's form against AndroidAPS's, and the published fit against the values in the abstract and on the poster's figure. It also checks the ways of reading the curve, and that the refit returns a quartic it was given.

Figures from other analyses are quoted from where they were produced. The measurement on other cohorts is in `results/inv012_peak_trough.md` and section 7 of document 5. The adult figure of 1915 is from `docs/preprint-dosing-constants.md`, and the overnight test is in `oref-investigations/results/bridge_diabeloop.md`.

Earlier drafts of this document are in git. The first (commit fd3662e) proposed the profile-anchored, flat form now kept as an arm. The second (commit b324528) opened on the comparison with AndroidAPS and described both systems for a general reader.

Reference

1. Romero-Ugalde HM, Adenis A, Cartier-Michaud T, Gauthier P, Desir C, Le Roux-Mallouf T, Huneker E, Benhamou PY. 920-P: The Relationship between Glycemia and Insulin Sensitivity Factor. Diabetes 2025;74(Supplement 1):920-P. doi:10.2337/db25-920-P